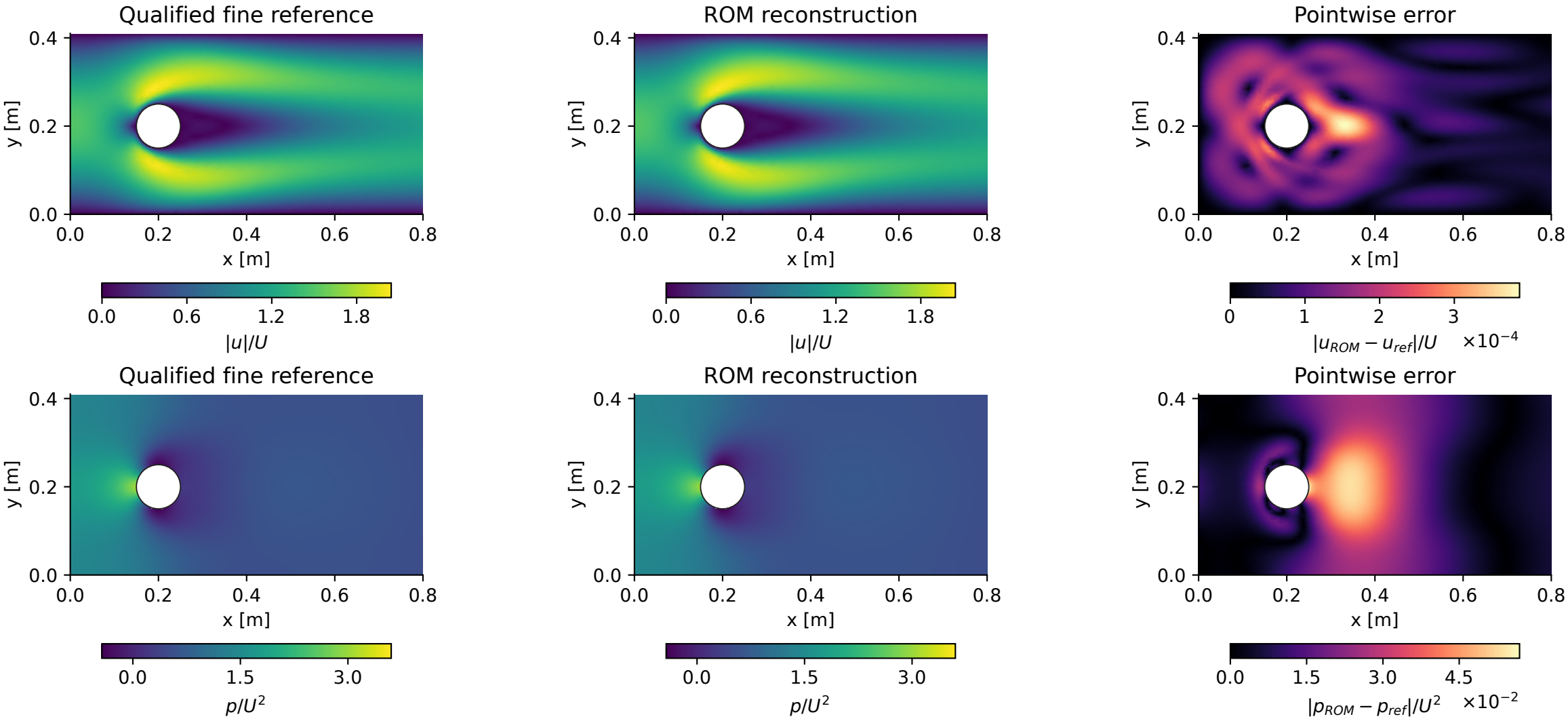

Standard POD–Galerkin: solved flow — Re = 23.125
Velocity (top) and pressure (bottom); near-cylinder view
Qualified fine reference
ROM reconstruction
Pointwise error
y [m]
x [m]
0.0
0.2
0.4
0.6
0.8
0.0
0.6
1.2
1.8
|u|/U
0
1
2
3
$|u_{ROM} - u_{ref}|/U$
$\times 10^{-4}$
Qualified fine reference
ROM reconstruction
Pointwise error
0.0
1.5
3.0
4.5
p/U^2
$|p_{ROM} - p_{ref}|/U^2$
$\times 10^{-2}$
Two POD velocity modes plus pressure-stability enrichment. Actual converged flow solve.
Display interpolation of saved samples; error scales differ between figures. Development cases, not validation.